Medical Diagnostic Retrieval System

Milestone 2 — Dataset & Queries

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Track: A — RAG Pipeline **Engine:** PostgreSQL 16

GitHub: https://github.com/Arunyaiitm/medical-diagnostic-retrieval-system

1. Data Source

Primary dataset: NIH Chest X-Ray Dataset (Sample) — publicly available on Kaggle.

- **Link:** https://www.kaggle.com/datasets/nih-chest-xrays/sample
- **File:** `sample_labels.csv` (5,606 records)
- **License:** Public domain (NIH Clinical Center)

Supporting data: Generated by `seed_data.py`, a documented Python script that creates realistic hospital data around the real NIH records. Constraints include specialisation-to-diagnosis mapping, chronological visit ordering from follow-up numbers, and real medication-to-condition pairing (e.g. Pneumonia → Azithromycin 500mg). Severity is derived from co-occurring findings: 1 finding = MILD, 2 = MODERATE, 3+ = SEVERE.

2. Database Statistics

Table	Rows	Data Source
departments	5	Seed script
doctors	30	Seed script
patients	4,230	NIH dataset
visits	5,606	NIH + seed
scans	5,606	NIH dataset
diagnoses	6,978	NIH dataset
prescriptions	3,934	Seed script
Total	**26,389**	**Well above Track A minimum of 1,000**

3. Data Dictionary

3.1. sample_labels.csv (NIH — real data)

Column	Type	Example	Description / Allowed Range
Image Index	string	00000013_005.png	Unique X-ray image filename
Finding Labels	string	Pneumonia—Infiltration	Pipe-separated disease labels
Follow-up #	int	5	0 = initial visit, 1+ = follow-up
Patient ID	int	13	Unique patient identifier
Patient Age	string	060Y, 013M	Y = years, M = months, D = days
Patient Gender	char	M	M (male) or F (female)
View Position	string	PA	PA or AP
OriginalImageWidth	int	3056	Image width in pixels
OriginalImageHeight	int	2544	Image height in pixels
PixelSpacing_x	float	0.139	Millimetres per pixel

14 disease labels: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural_Thickening, Pneumonia, Pneumothorax, No Finding.

3.2. Key database columns after seeding

patients: `patient_id` (INTEGER, from NIH), `age` (INTEGER, 0–130, parsed from formats like 060Y / 013M), `gender` (CHAR(1), M/F), `blood_group` (VARCHAR(5), realistic distribution: A+, B+, O+, AB+ etc.)

visits: `visit_date` (TIMESTAMP, derived from follow-up number), `visit_type` (Outpatient / Inpatient / Emergency / Follow-up), `chief_complaint` (TEXT, realistic medical complaints)

scans: `image_filename` (VARCHAR(255), UNIQUE, from NIH), `view_position` (PA/AP), `image_width / height` (INTEGER), `pixel_spacing` (NUMERIC(5,3))

diagnoses: `finding_label` (VARCHAR(100), one of 14 NIH labels + No Finding), `severity` (MILD / MODERATE / SEVERE), `notes` (TEXT, template-based clinical descriptions)

prescriptions: `medication_name` (VARCHAR(150), real drugs), `dosage` (VARCHAR(50), e.g. 500mg), `frequency` (VARCHAR(50)), `duration_days` (INTEGER, 1–365)

doctors: `specialization` (Radiologist / Pulmonologist / Cardiologist / General Physician / Emergency Medicine), `department_id` (FK → departments, matched by specialisation)

4. Import Steps

Prerequisites: PostgreSQL 16 installed, Python 3.10+, `pip install psycopg2-binary`

1. **Create database:**
   `psql -U postgres -c "CREATE DATABASE medical_rag;"`
2. **Run schema:**
   `psql -U postgres -d medical_rag -f schema.sql`
   Expected output: 7× `CREATE TABLE` + 10× `CREATE INDEX`
3. **Download dataset:**
   Get `sample_labels.csv` from Kaggle (link in Section 1). Place in the same folder as `seed_data.py`.
4. **Configure password:**
   Open `seed_data.py`, change `DB_PASSWORD = "your_password"` to your actual postgres password.
5. **Seed the database:**
   `cd data/ && python seed_data.py`
   Expected: 26,389 rows across 7 tables (see Section 2 for breakdown)
6. **Run queries:**
   `psql -U postgres -d medical_rag -f queries.sql`
   Expected: All 10 queries return results with zero errors

5. Queries (`queries.sql`)

#	Type	Description
1	Aggregation	Count of diagnoses per disease type (GROUP BY + ORDER BY)
2	Aggregation	Average patient age per department (AVG across 4 joins)
3	Join (4 tables)	Full history: patient → visit → scan → diagnosis → prescription
4	Join (3 tables)	Doctors and diagnosis counts grouped by department
5	Subquery	Patients with more diagnoses than average (nested subquery)
6	Subquery	Scans with no diagnosis yet (NOT IN subquery)
7	CTE	Patient diagnostic journey timeline (WITH + HAVING)
8	CTE	Monthly diagnosis trends per disease (DATE_TRUNC)
9	Window (RANK)	Rank doctors by diagnoses within each department
10	Window (ROW_NUMBER)	Running total of visits per patient over time

Coverage: 2 aggregations, 2 joins, 2 subqueries, 2 CTEs, 2 window functions — all required types met.

6. Submission Contents

- `sample_labels.csv` — NIH dataset (5,606 rows of real data)
- `seed_data.py` — documented Python seed script with realistic constraints
- `queries.sql` — 10 labelled queries covering all required types
- `README.md` — data dictionary, import steps, query descriptions